

Gordon Bie

D'Amore McKim School of Business, Northeastern University, Boston, MA

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RESEARCH INTERESTS

- Operations Decisions: causal ML for product rollouts with spillovers, delayed feedback, & fair customer service design
- Dynamic Marketplaces: matching & pricing on platforms when participants value jobs differently & act strategically
- Collective Intelligence: when human-AI collaboration beats delegation & whether mediation helps teams decide better
- Agentic AI & Human Capital: whether AI gains require workflow redesign & how they reshape worker skills & roles

EDUCATION

Northeastern University

2023 - 2027

B.S. in International Business, Concentration: Business Analytics, Minor: Data Science

GPA: 3.74/4.00

Independent Studies

2025 - 2026

Data Mining & Kernelized Machine Learning (Prof. Chieh Wu), Statistical Inference (Prof. Chieh Wu), Deep Learning Fundamentals (DeepLearning AI), ML in Production (DeepLearning AI), Real Analysis (18.100A MIT OpenCourseWare)

RESEARCH EXPERIENCE

Northeastern University College of Engineering

Boston, MA

Research Assistant

2025 - Present

- Worked as research assistant under Prof Xiaoning “Sarah” Jin and Zifeng Wang in industrial engineering research
- Reviewed two decades of time-series tokenization literature, from symbolic encodings to LLM-based forecasters
- Synthesized existing methods into a 2×2 taxonomy: discrete/continuous tokens × fixed/adaptive segmentation
- Designed controlled experiment testing different tokenizer types per quadrant: SAX, ABBA, PatchTST, Chronos
- Evaluated signal reconstruction & downstream prediction separately on cross-sectional & time-series data
- Discovered that better reconstruction doesn't guarantee accuracy: SAX beat PatchTST despite 140× higher error
- Applying findings to design a novel tokenization technique & evaluation for cyclic multivariate machine signals

Skinapse Labs

Boston, MA

Research Scientist

2024 - Present

- As founding member of technical staff of AI dermatological lab startup led classification ML model development
- Designed multinomial models estimating $P(\text{neighbor class} \mid \text{source class, distance})$ over 46.6K acne lesions
- Built validation around image-grouped 75/25 splits & 1,000 image-level bootstrap resamples for 95% CIs
- Showed all 6 lesion classes cluster beyond permutation null ($p \approx 0.001$), yielding spatial priors for CNNs
- Specified a Bayesian network layer combining raw CNN confidence with spatial priors into posterior scores
- Directed a multi-university intern team, assigning lesion-pair configurations & per-model analysis plans
- Designing a pipeline that uses a multimodal LLM to extract structured skin-color features from facial acne images
- Training a random forest classifier on LLM-extracted color features to predict facial acne severity grades
- Testing whether color cues carry severity signal, as a lightweight complement to CNN-based lesion detection

INDUSTRY EXPERIENCE

Flow Traders

New York, NY

Software Engineer Intern

2026 - Present

- Building infrastructure code, data & CI/CD pipelines for HFT trading using Python, Kubernetes, Ansible & GHA
- Designing episodic memory & output evaluation framework for trading engine health diagnostics AI service agent
- Created global cross enterprise Atlassian BitBucket to Github Enterprise repository migration framework & script
- Added live monitoring & predictive memory adjustment for Java trading applications across distributed systems

Cortex	Boston, MA
<i>Machine Learning Engineer Intern</i>	2026 - 2026
• Worked with 8 engineers to develop internal AI knowledge graph tool in Python & Typescript with Docker & Git • Architected novel GraphRAG pipeline combining Neo4j, pgvector embeddings, NER & cosine similarity search • Built pipelines parsing PDFs & CSVs into structured JSON stored across Neo4j, S3 & Supabase PostgreSQL • Reduced hardware engineering design research process by 99% with natural language knowledge graph search	
State Street Corporation	Boston, MA
<i>Artificial Intelligence Engineer Intern</i>	2025 - 2025
• Delivered a full-stack AI app using LLM Agents, FastAPI, & React to automate GAAP-compliant audit reports • Built REST API routers & data pipelines transforming structured data from Databricks to enable LLM automation • Implemented prompt evaluation in MLflow testing 10+ prompts via cos_sim, multi-LLM critique & auditor input • Reduced audit report preparation from 2-3 weeks to 1 day, saving hundreds of hours for global risk & audit teams	
Northeastern University Sherman Center for Engineering	Boston, MA
<i>Data Scientist Intern</i>	2025 - 2025
• Architected FastAPI app linking Slack, Jira & Trello APIs with SBERT embeddings & pgvector similarity search • Built RAG-based agentic deduplication with Gemini reasoning, achieving 95% precision on redundant tasks • Deployed app on Streamlit, PostgreSQL & Docker microservices, cutting product team task redundancy by 30%+	
State Street Investment Management	Boston, MA
<i>Client Experience Analyst Intern</i>	2025 - 2025
• Drove retention of a major institutional client via tailored sentiment decks delivered to sales execs & leadership • Built stakeholder-specific decks across 2000+ clients in various sectors, presenting findings & actionable insights • Surfaced recurring client themes from feedback data, informing go-forward sales strategy & account prioritization	
Apple Inc	Remote
<i>Marketing Analytics Extern</i>	2024 - 2024
• Analyzed 50K+ survey responses in Python, reducing data processing time by 75% for consumer audio teams • Built Plotly dashboards visualizing 15 consumer behaviour metrics & hastening product decision-making cycles • Developed NLP sentiment pipeline processing 200+ news articles/week with 89% accuracy for market insights	
Indigo Education	Vancouver, BC
<i>CRM & Sales Analyst Intern</i>	2023 - 2025
• Led customer support initiatives at the education agency, increased client satisfaction & retention by 60% • Secured 20 new clients via digital sales & in-person pitch meetings increasing firm revenue by ~\$600,000USD • Provided training for 5 incoming CRM associates, trained them to ensure success in client service & retention	

HONORS & AWARDS

D'Amore McKim School of Business at Northeastern University	2024 - 2026
<i>Dean's List Honors</i>	

SCHOLARSHIPS & FELLOWSHIPS

<i>D'Amore McKim School of Business Scholarship</i>	2026
<i>PEAK Basecamp Research Fellowship</i>	2026
<i>Ray Kinnunen Scholarship</i>	2025
<i>Judith Memorial Pre-Law Scholarship</i>	2023

POSTER PRESENTATIONS

Bie, G., & Jin, X. 2026. *Applying NLP techniques to high-frequency multivariate time-series data in manufacturing machinery* [Poster presentation]. Research, Innovation, Scholarship & Entrepreneurship (RISE) Expo, Northeastern University, Boston, MA, United States.

Nguyen, J. [John], Shah, E., Simeone, J., Kim, A., **Bie, G.**, & Nguyen, J. [Josephine]. (under review). 2026. *A multinomial and permutation-based framework for modeling spatial relationships among acne lesion classes* [Abstract submitted for poster presentation]. Southeastern Medical Scientist Symposium.

Nguyen, J. [John], Shah, E., Simeone, J., Kim, A., **Bie, G.**, & Nguyen, J. [Josephine]. (under review). 2026. *Spatial Neighborhood Patterns as Contextual Features for Acne Lesion Classification*. [Abstract submitted for poster presentation]. American Academy of Dermatology.

TEACHING EXPERIENCE

Northeastern University

Boston, MA

Calculus Tutor

2024 - Present

- Tutored 15 undergraduates in differential equations, advanced integration, multivariable calculus & limits
- Designed a calculus curriculum of practice problem sets & mini-quizzes to gauge understanding by topic
- Created custom study plans for each student, tailoring pacing & practice problems to individual gaps
- Ran review sessions before midterms & finals to prepare students with exam-style practice problems
- Coached all 15 students to an A in their courses by making trying to make calculus engaging instead of a chore

Indigo Education

Remote

Highschool Math Tutor

2023 - Present

- Tutored grade 10–12 students remotely in IB Mathematics: Analysis & Approaches HL & AP Calculus AB/BC
- Prepared students for IB Math HL & AP Calculus exams, adapting lessons to each curriculum's syllabus
- Coached students to 5s on AP Calculus AB/BC, 6s & 7s on IB Math AA HL, IB's two highest grades
- Prioritized students' enjoyment & appreciation of mathematics over grades alone, fostering lasting interest
- Mentored 30 highschool students who went on to study engineering & mathematics at top universities worldwide

Northeastern University

Boston, MA

Statistics Tutor

2024 - 2025

- Tutored 10 students in conditional probability, distributions, hypothesis testing, regression & inferential statistics
- Developed a statistics curriculum with practice problem sets & mini-quizzes to check mastery by topic
- Built individualized study plans that adapted pacing & practice to each student's weak topics & scheduling quirks
- Ran review sessions before midterms & finals to prepare students with exam-style practice problems
- Guided all 10 students to an A in their courses, but more importantly helped students cultivate interest in statistics

Indigo Education

Remote

Highschool Analytical Writing Tutor

2023 - 2025

- Taught analytical writing & reading comprehension, from interpreting themes to spotting rhetorical devices
- Trained students to build logically structured arguments with clear theses, smooth transitions & textual evidence
- Strengthened students' grammar & syntax, refining sentence structure for clarity & precision in written arguments
- Helped students uncover the beauty in literature, fostering genuine appreciation for literature beyond the syllabus

EXTRACURRICULARS

D'Amore McKim School of Business at Northeastern University

Boston, MA

Bachelor of Science in International Business Student Mentor

2026 - Present

- Mentoring undergraduates in Northeastern's BSIB program, advising on course planning, career & studies abroad

Global Research & Consulting Group*Junior Associate Consultant***Boston, MA***2024 - 2025*

- Delivered strategic analysis & operational recommendations to international organizations & public-sector clients

Disrupt Fintech Society*Marketing Coordinator***Boston, MA***2024 - 2025*

- Planned, marketed & executed events for students in fintech society such as hackathons, panels & research talks

Generate Product Development Studio*Data Science Technical Lead***Boston, MA***2025 - 2026*

- Technical lead for student led product development studio working on ML & AI software products for startups

Office of Global Experience at Northeastern University*Global Program Alumni Ambassador***Boston, MA***2024 - 2025*

- Planned & presented about the global experience programs at Northeastern for prospective & incoming students

SKILLS & INTERESTS

Programming & Query Languages: Python, R, SQL, JavaScript, TypeScript, CSS/HTML, Racket, Cypher**Frameworks & Libraries:** Sklearn, TensorFlow, PyTorch, Jax, SciPy, NumPy, Pandas, Seaborn, FastAPI, React, Node.js**Tools:** Git, Linux, Powershell, Claude Code, Ansible, AWX, Docker, Kubernetes, ArgoCD, Airflow, Prometheus, Grafana**Databases:** MySQL, PostgreSQL/pgvector, SQLite, MongoDB, Neo4j, Kuzu, AWS S3, OracleSQL 12**Interests:** Burgers, Chess, Poker (PLO & NLTH), Geography, Horology, Pop Music (Keyboard & Vocal), Textiles**Natural Languages:** Native Mandarin, Professional Spanish, Intermediate French, Intermediate Japanese**REFERENCES**

Dr. Barbara Z Larson*Email: b.larson@northeastern.edu***Executive Professor of Management, Northeastern University****Dr. Chieh Wu***Email: ch.wu@northeastern.edu***Associate Professor, Northeastern University****Dr. Xiaoning “Sarah” Jin***Email: xi.jin@northeastern.edu***Associate Professor, Northeastern University****Dr. Eric Bogert***Email: e.bogert@northeastern.edu***Assistant Professor, Northeastern University****Dr. Daehan Yu***Email: d.yu@northeastern.edu***Assistant Professor, Northeastern University**